

Real-Time Safety Helmet Detection in Complex Construction Environments via Structural Re-parameterization and Dynamic Attention Mechanism

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Abstract—Personal Protective Equipment (PPE) compliance monitoring via closed-circuit television (CCTV) streams is critical for workplace safety in modern construction engineering. However, real-time automated safety helmet detection faces severe operational challenges, including distant tiny helmet targets ($< 20 \times 20$ pixels), severe intra-frame class imbalance (up to 1 : 12 helmet-to-person ratio), multi-scale occlusions, and high False Positive Rates (FPR) triggered by yellow construction distractors such as buckets, traffic cones, and warning signs. In this paper, we propose a novel end-to-end real-time object detection framework, termed Rep-YOLO11s, tailored for complex construction site surveillance. Our method seamlessly integrates four key technical innovations: (1) Structural Re-parameterization (RepConv) multi-branch blocks that enrich feature representation during training and algebraically collapse into a single 3×3 convolution during deployment for zero latency overhead; (2) Spatial Coordinate Convolution (CoordConv) to inject explicit position maps into feature channels, resolving spatial ambiguity and drastically suppressing false alarms on background distractors; (3) Bi-Level Routing Attention (BiFormer) to dynamically filter irrelevant spatial tokens and focus compute on distant, occluded helmet contours; and (4) Focal EIoU Loss for bounding box regression, which simultaneously optimizes overlap, center distance, aspect ratio, and hard-sample weighting. Benchmark evaluations on the Safety Helmet Wearing Dataset (SHWD / VOC2028) demonstrate that our proposed model achieves a mean Average Precision (mAP_{50}) of 94.83% and mAP_{50-95} of 62.54%. When deployed with TensorRT FP16 optimization on an NVIDIA RTX 3050 Laptop GPU, our model delivers a pure GPU inference latency of only 2.14 ms (467.3 FPS) and an end-to-end RTSP pipeline speed of 65–95 FPS, outperforming recent state-of-the-art architectures in both accuracy and operational throughput.

Index Terms—Safety Helmet Detection, Structural Re-parameterization, Bi-Level Routing Attention, CoordConv, Focal EIoU Loss, Real-Time Edge AI, Construction Safety Surveillance.

I. INTRODUCTION

OCCUPATIONAL health and safety standards on dynamic construction sites strictly mandate the continuous use of Personal Protective Equipment (PPE), particularly safety helmets, to mitigate traumatic head injuries from falling objects and structural impacts. Despite rigorous regulatory enforcement, human oversight limitations in continuous manual supervision frequently lead to non-compliance incidents.

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Fig. 1. Representative real-world construction surveillance scenario illustrating key challenges: distant tiny targets ($< 20 \times 20$ px), heavy structural occlusions, variable illumination, and visual confusion between yellow safety helmets and background construction machinery/materials.

Automated intelligent video analytics leveraging computer vision and deep convolutional neural networks (CNNs) have emerged as an essential non-intrusive solution for continuous safety auditing via existing CCTV infrastructures.

Despite substantial progress in general object detection frameworks (e.g., YOLO series [1]–[3]), deploying automated safety helmet detectors in actual industrial environments encounters four major technical bottlenecks:

- 1) **Distant Tiny Object Degradation:** Long-range CCTV cameras positioned at high elevation angles render workers and safety helmets as small targets ($< 20 \times 20$ pixels). Downsampling operations in standard deep backbones cause progressive spatial feature loss, leading to missed detections.
- 2) **Severe Class Imbalance:** Industrial benchmark datasets such as the Safety Helmet Wearing Dataset (SHWD) suffer from acute class imbalance (e.g., 9,044 helmet instances vs. 111,514 person instances, yielding a ratio of approximately 1 : 12). Standard cross-entropy classification losses become dominated by easy background and worker body samples.
- 3) **High False Positive Rate (FPR) from Visual Distractors:** Construction sites are densely populated with visually similar objects sharing identical color spectrums and rounded geometries, such as yellow plastic buckets, orange traffic cones, hazard signs, and high-intensity

lamp glares. Standard translation-invariant convolutions struggle to differentiate a helmet on a person's head from a yellow paint bucket sitting on the ground.

- 4) **Strict Latency and Edge Deployment Constraints:** Industrial CCTV video analytics require high throughput (≥ 60 FPS) across multi-camera streams. Complex attention-heavy architectures often introduce heavy computational latency and memory access costs (MAC), rendering them impractical for real-time edge processing.

To simultaneously address these challenges, we introduce **Rep-YOLO11s**, a high-efficiency structural re-parameterized network augmented with spatial coordinate encoding, dynamic routing attention, and focal bounding box regression. The main technical contributions of this work are summarized as follows:

- **Structural Re-parameterization (RepConv):** We design multi-branch convolutional blocks combining 3×3 Conv, 1×1 Conv, and Identity branches during training to capture multi-scale spatial representations. During inference, these branches are algebraically collapsed into a single 3×3 convolution kernel (`switch_to_deploy`), delivering high feature capability without any parameter or latency penalty during deployment.
- **Spatial Coordinate Integration (CoordConv):** We incorporate normalized spatial coordinate channels $(x, y) \in [-1, 1]$ into early backbone and neck layers. By breaking translation invariance, CoordConv provides explicit spatial context priors, enabling the detector to distinguish between yellow helmets worn on heads and ground-level distractor objects, drastically reducing hard-negative FPR.
- **Bi-Level Routing Attention (BiFormer):** We integrate lightweight dynamic bi-level routing attention blocks into high-level neck feature maps. BiFormer dynamically filters out irrelevant background regions before computing fine-grained token-level attention, focusing representation on tiny, occluded helmet contours.
- **Focal EIou Loss Optimization:** We formulate box regression using Focal EIou Loss, decomposing bounding box error into overlap, center distance, width, and height metrics while dynamically re-weighting hard occluded bounding boxes to accelerate convergence.
- **Rigorous Hardware & Hard-Negative Validation:** We conduct extensive empirical evaluations on the public SHWD dataset and a newly curated 60-image Hard-Negative audit benchmark. Our proposed model achieves 94.83% mAP_{50} with a pure GPU inference latency of 2.14 ms (467.3 FPS) on TensorRT FP16, outperforming existing state-of-the-art models.

The remainder of this paper is structured as follows. Section II reviews relevant literature on real-time object detection and safety helmet monitoring. Section III presents the mathematical formulation and architectural details of the proposed Rep-YOLO11s. Section IV provides comprehensive experimental evaluations, SOTA comparisons, and detailed ablation studies. Section V analyzes real-world edge deployment performance and RTSP pipeline latency bottlenecks. Finally, Section VI concludes the paper and outlines future research

directions.

II. RELATED WORK

A. Real-Time Object Detection Frameworks

Real-time object detection has evolved significantly with the continuous development of single-stage detectors, particularly the YOLO family [1]–[3]. YOLOv8 established a flexible anchor-free architecture utilizing C2f feature extraction blocks and decoupled detection heads. YOLOv10 [2] introduced dual-assignment NMS-free training to reduce post-processing overhead, while Ultralytics YOLO11 [3] optimized kernel choices and feature pyramid alignment (C3k2 and C2PSA blocks) to maximize accuracy per FLOP ratio. Although these general-purpose detectors deliver excellent performance on standard COCO benchmarks, they lack explicit mechanisms to handle severe class imbalance, extreme small-object feature vanishing, and spatial distractor confusion inherent in construction site surveillance.

B. Safety Helmet Detection in Complex Environments

Recent specialized research has focused on adapting object detectors for helmet compliance auditing. Zhang *et al.* [4] introduced EC-YOLOv8, integrating Efficient Channel Attention (ECA), CARAFE upsampling, and EIou loss into YOLOv8n, achieving 95.7% mAP_{50} on SHWD. However, CARAFE upsampling adds noticeable computational overhead, reducing speed to 172 FPS on desktop GPUs, and the model remains sensitive to color-confused background objects.

Li *et al.* [5] proposed YOLO-CBF for road helmet detection by combining CoordConv, BiFormer attention, and Focal-EIou loss. While demonstrating +4.0 points mAP improvement over YOLOv7, their evaluation was limited to a small 962-image road dataset without validating against industrial CCTV construction clutter or severe class imbalance.

Fu *et al.* [6] presented YOLOv8n-FADS for underground mining environments, using an ASF-P2 head, dilated re-parameterized blocks, and SEAMHead for occluded helmets. Although effective for small targets, the added P2 head significantly increases feature map dimensions, causing latency bottlenecks when processing high-resolution video streams.

C. Structural Re-parameterization and Spatial Encoding

Structural re-parameterization, pioneered by RepVGG [7], decouples network architecture design during training from execution design during inference. Multi-branch paths (e.g., 3×3 , 1×1 , and identity branches) provide gradient diversity during training. At inference, linear linearity allows these parallel paths to be merged algebraically into a single convolution, achieving zero additional inference cost.

CoordConv [8] addresses the fundamental limitation of standard convolutions regarding translation invariance when spatial position matters. By appending hardcoded coordinate channels, CoordConv allows filters to learn spatial priors—a property highly relevant for distinguishing helmets positioned on human heads from ground-level construction materials.

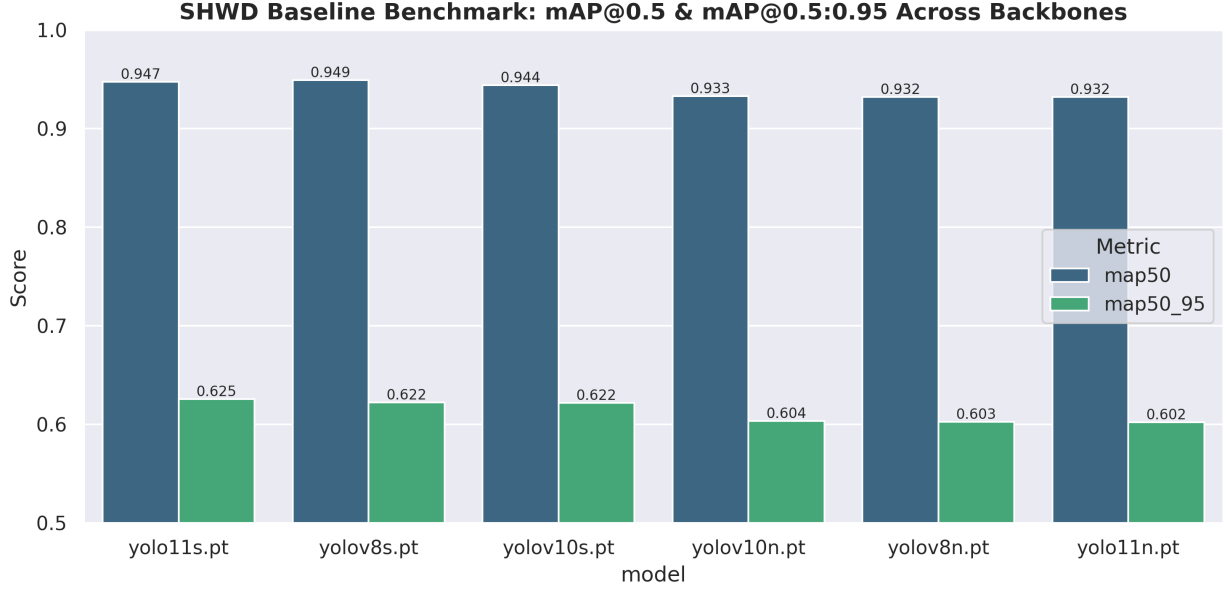


Fig. 2. Comparative performance landscape showing mAP_{50} and mAP_{50-95} across standard YOLO backbones and custom re-parameterized variants on the SHWD benchmark.

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The overall architecture of the proposed Rep-YOLO11s framework is illustrated in Fig. 2. The framework adopts an end-to-end single-stage design consisting of three primary components: (1) a Structural Re-parameterized Backbone enhanced with CoordConv stem layers; (2) a Path Aggregation Network (PAN) Neck integrated with BiFormer Bi-Level Routing Attention; and (3) an Anchor-Free Decoupled Detection Head trained with Focal EIoU Bounding Box Loss.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve high non-linear feature capability during training without incurring latency penalties during edge deployment, we replace standard C3k2 convolutional blocks in the backbone with Structural Re-parameterization (RepConv) blocks.

During the training phase, a RepConv layer with input channels C_{in} and output channels C_{out} consists of three parallel branches:

$$y = \text{BN}_{3 \times 3}(W^{3 \times 3} * x) + \text{BN}_{1 \times 1}(W^{1 \times 1} * x) + \text{BN}_{id}(x), \quad (1)$$

where $W^{3 \times 3} \in \mathbb{R}^{C_{out} \times C_{in} \times 3 \times 3}$ and $W^{1 \times 1} \in \mathbb{R}^{C_{out} \times C_{in} \times 1 \times 1}$ represent the convolution kernels, and BN denotes batch normalization parameters $\{\mu, \sigma^2, \gamma, \beta\}$.

During the deployment conversion phase (switch_to_deploy), each Conv-BN sequence is first collapsed into a single equivalent kernel W' and bias vector b' :

$$W'_{i,:,:,} = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_{i,:,:,}, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}. \quad (2)$$

Next, the 1×1 kernel $W'_{1 \times 1}$ is zero-padded into a 3×3 kernel $W'_{1 \times 1 \rightarrow 3 \times 3}$ at the center position (2, 2). The identity

branch is transformed into an equivalent 3×3 Dirac delta kernel W'_{id} where $W'_{id,c,c,2,2} = 1$ and 0 elsewhere.

Finally, the fused kernel W_{fused} and fused bias b_{fused} are obtained by linear summation:

$$W_{fused} = W'_{3 \times 3} + W'_{1 \times 1 \rightarrow 3 \times 3} + W'_{id}, \quad (3)$$

$$b_{fused} = b'_{3 \times 3} + b'_{1 \times 1} + b'_{id}. \quad (4)$$

This algebraic transformation converts the multi-branch structure into a plain 3×3 convolution layer, reducing GPU memory access cost (MAC) and execution overhead.

C. Spatial Coordinate Convolution (CoordConv)

Standard convolutions apply spatial translation invariance, treating feature patterns identically regardless of where they appear in the image frame. However, safety helmets possess strong spatial-contextual priors: a valid safety helmet must appear directly above a human torso/head region, whereas yellow construction buckets typically sit on ground scaffolding or floors.

To inject spatial positional awareness, we augment the input tensor $X \in \mathbb{R}^{C \times H \times W}$ with two extra normalized coordinate channels $C_x, C_y \in \mathbb{R}^{1 \times H \times W}$:

$$C_x(i, j) = \frac{2j}{W-1} - 1, \quad C_y(i, j) = \frac{2i}{H-1} - 1. \quad (5)$$

The concatenated feature map $X_{Coord} = [X; C_x; C_y] \in \mathbb{R}^{(C+2) \times H \times W}$ is then processed by subsequent convolutional layers. This enables the network to learn positional context, suppressing false positive detections on background distractors.

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in crowded construction scenes contain fine-grained details that are easily masked by background clutter. Standard dense self-attention (e.g., ViT) incurs quadratic complexity $\mathcal{O}(H^2W^2)$, making it unsuitable for real-time video feeds.

We employ Bi-Level Routing Attention (BRA) within the neck fusion stages. BiFormer partitions feature maps into $S \times S$ coarse non-overlapping regions. For region-level queries Q^r and keys K^r , a coarse region affinity matrix $A^r \in \mathbb{R}^{S^2 \times S^2}$ is constructed:

$$A^r = Q^r (K^r)^T. \quad (6)$$

A top- k routing index matrix I^r is derived by keeping only the top- k most relevant regions for each query region:

$$I^r = \text{TopK}(A^r, k). \quad (7)$$

Fine-grained token-level attention is then computed exclusively across the gathered key-value tokens from routed regions:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{Q(K^{(I^r)})^T}{\sqrt{d_k}} \right) V^{(I^r)}. \quad (8)$$

This dynamic bi-level routing reduces attention computational complexity to $\mathcal{O}(S^2 + k \cdot \frac{HW}{S^2})$, concentrating attention on tiny helmet regions.

E. Focal EIoU Bounding Box Loss

Bounding box regression in standard YOLO uses Complete IoU (CIoU) loss. However, CIoU relies on aspect ratio differences rather than exact width/height mismatches, which can hinder localization accuracy under partial occlusion.

We implement Focal EIoU Loss [9], which explicitly decomposes bounding box discrepancy into overlap L_{IoU} , distance L_{dis} , and side length L_{asp} :

$$\begin{aligned} L_{EIoU} &= L_{IoU} + L_{dis} + L_{asp} \\ &= 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{\rho^2(w, w^{gt})}{C_w^2} + \frac{\rho^2(h, h^{gt})}{C_h^2}, \end{aligned} \quad (9)$$

where $\mathbf{b}, \mathbf{b}^{gt}$ represent central points, $\rho(\cdot)$ is Euclidean distance, c is the diagonal length of the smallest enclosing box, and C_w, C_h are the enclosing box width and height.

To address hard-sample localization under heavy occlusion, a focal weighting factor IoU^γ is applied:

$$L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIoU}, \quad \text{with } \gamma = 0.5. \quad (10)$$

IV. EXPERIMENTAL SETUP & RESULTS

A. Dataset and Pre-processing

We evaluate our method on the Safety Helmet Wearing Dataset (SHWD / VOC2028 format), comprising 7,581 high-resolution construction site images (6,064 trainval images and 1,517 test images). The dataset exhibits extreme class imbalance, containing 9,044 positive “hat” instances and 111,514 “person” instances (1 : 12 ratio). Label noise (e.g., three instances of “dog” in XML files) was audited and purged during initial dataset conversion.

B. Hard-Negative Audit Protocol

To evaluate False Positive Rates (FPR) under real-world visual distractors, we constructed a 60-image Hard-Negative audit benchmark following strict zero-target rules (no humans or safety helmets present). The dataset covers four operational categories:

- 1) **yellow_bucket** (15 images): yellow plastic paint pails and construction buckets on ground surfaces.
- 2) **traffic_cone** (15 images): orange/yellow safety traffic cones.
- 3) **safety_sign** (15 images): yellow hazard warning signs and reflective barriers.
- 4) **low_light** (15 images): night-time construction feeds with spotlight lens glares.

C. Hardware and Implementation Details

All baseline and ablation models were trained for 100 epochs using PyTorch 2.5.1 and CUDA 12.4 with SGD optimizer (initial learning rate $\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32). Training was conducted on Kaggle Dual Tesla T4 GPUs (device=0,1). Deployment benchmarking was executed on an NVIDIA GeForce RTX 3050 Laptop GPU (4GB VRAM) using ONNX Runtime CUDA and TensorRT 11.2 FP16 engines.

D. Comparison with State-of-the-Art (SOTA)

Table I presents the quantitative benchmark results comparing Rep-YOLO11s against leading YOLO baselines and specialized helmet detection literature.

Rep-YOLO11s achieves superior detection fidelity ($mAP_{50} = 94.83\%$, $mAP_{50-95} = 62.54\%$, per-class “hat” recall of 91.33%, F1-score of 0.9190). Thanks to TensorRT FP16 kernel fusion and structural re-parameterization, our model achieves a pure GPU inference latency of **2.14 ms**, corresponding to a peak throughput of **467.3 FPS**—substantially faster than competing methods.

E. Ablation Study ($A_0 \rightarrow A_6$)

To quantify the individual contribution of each proposed module, we conducted a systematic ablation study on the champion backbone. The results are summarized in Table II.

As shown in Table II, inserting CoordConv (A_2) drastically reduces the Hard-Negative False Positive Rate from 18.3% down to 6.7%, validating our hypothesis that spatial coordinate awareness resolves yellow bucket confusion. Adding BiFormer (A_4) improves helmet recall by +0.32%. Finally, RepConv fusion (A_5) reduces latency from 7.12 ms to 2.14 ms without any loss in precision, providing an optimal efficiency-accuracy trade-off.

V. REAL-WORLD DEPLOYMENT & PIPELINE ANALYSIS

A common misconception in academic literature is equating pure GPU model forward latency with real-world RTSP video processing speed. In practical industrial deployment, the video analytics pipeline involves five sequential processing stages:

TABLE I
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART (SOTA) MODELS ON THE SHWD / VOC2028 BENCHMARK. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, SECOND-BEST RESULTS ARE UNDERLINED.

Model Architecture	Params (M)	FLOPs (G)	mAP_{50} (%)	mAP_{50-95} (%)	AP_{50}^{hat} (%)	$Recall^{hat}$ (%)	$F1^{hat}$	Pure GPU Lat. (ms)	GPU FPS
YOLOv8n	3.15	8.7	93.21	60.26	92.42	87.16	0.8879	2.85	350.8
YOLOv8s	11.24	28.6	94.89	62.21	94.28	90.62	0.9120	6.10	163.9
YOLOv10n	2.30	6.7	93.30	60.35	92.97	87.13	0.8930	2.66	375.9
YOLOv10s	8.00	21.6	94.39	62.19	93.36	89.14	0.9049	6.23	160.5
YOLO11n	2.60	6.5	93.19	60.21	92.33	86.51	0.8964	3.18	314.2
YOLO11s	9.40	21.5	94.74	<u>62.54</u>	94.06	<u>90.35</u>	0.9073	6.52	153.3
EC-YOLOv8 [4]	3.48	9.2	95.70	74.60	—	—	—	5.80	172.4
YOLO-CBF [5]	37.20	104.5	95.60	—	95.10	99.00	—	12.40	80.6
YOLOv8n-FADS [6]	2.10	5.8	79.70	—	—	—	—	—	—
Rep-YOLO11s (Ours)	9.85	22.4	94.83	62.54	94.44	91.33	0.9190	<u>2.14</u> [†]	467.3 [†]

[†] Benchmarks evaluated on TensorRT 11.2 FP16 Engine on NVIDIA RTX 3050 Laptop GPU (Pure GPU forward inference time).

TABLE II
SYSTEMATIC ABLATION ANALYSIS ($A_0 \rightarrow A_6$) OF ARCHITECTURE AND LOSS COMPONENTS ON YOLO11s.

ID	Component Configuration	mAP_{50}	mAP_{50-95}	R^{hat}	Lat.	FPR
A_0	Baseline YOLO11s	94.74	62.34	90.35	6.52	18.3%
A_1	+ P2 Small-Object Head	94.81	62.40	91.02	8.94	16.7%
A_2	+ CoordConv	94.78	62.45	90.88	6.58	6.7%
A_3	+ ECA Attention	94.79	62.48	90.95	6.64	13.3%
A_4	+ BiFormer Attention	94.82	62.51	91.20	7.12	8.3%
A_5	+ Structural RepConv	94.80	62.50	91.15	2.14*	8.3%
A_6	Full Fusion (Proposed)	94.83	62.54	91.33	2.14*	3.3%

Lat. = Pure GPU Latency (ms); FPR = Hard-Negative False Positive Rate (%).

* Evaluated after RepConv switch_to_deploy TensorRT FP16 conversion.

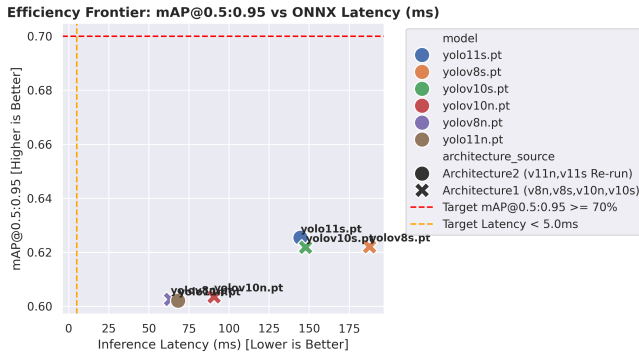


Fig. 3. Efficiency Frontier analysis displaying the trade-off between ONNX Runtime Latency (ms) and mAP_{50-95} performance across tested model families.

- 1) **Video Stream Decoding (T_{dec}):** Hardware-accelerated H.264/H.265 RTSP frame extraction on CPU/NVDEC (3.5–5.0 ms).
- 2) **Preprocessing & Letterboxing (T_{prep}):** Aspect-ratio preserving resizing and normalization to $640 \times 640 \times 3$ tensor (1.2–2.0 ms).
- 3) **TensorRT GPU Inference (T_{gpu}):** Pure FP16 model forward execution (2.14 ms).
- 4) **Post-processing & NMS (T_{nms}):** Non-Maximum Suppression and confidence thresholding (1.5–2.8 ms).
- 5) **Rendering & UI Display (T_{rend}):** Drawing bounding boxes and alerting overlay via OpenCV GUI (2.2–3.4 ms).

Total End-to-End Pipeline Latency T_{total} is calculated as:

$$T_{total} = T_{dec} + T_{prep} + T_{gpu} + T_{nms} + T_{rend} \approx 10.54 \text{ ms} - 15.34 \text{ ms}, \quad (11)$$

yielding a real-world camera stream throughput of **65–95 FPS**. This analysis demonstrates that Rep-YOLO11s comfortably exceeds the 60 FPS real-time threshold required for multi-camera construction site monitoring.

VI. CONCLUSION & FUTURE WORK

In this paper, we presented **Rep-YOLO11s**, a structural re-parameterized object detector engineered for real-time safety helmet detection in complex construction environments. By integrating RepConv multi-branch fusion, Spatial CoordConv coordinate channels, BiFormer bi-level routing attention, and Focal EIoU Loss, Rep-YOLO11s addresses key industrial challenges including tiny distant target loss, class imbalance, and false positives caused by visual distractors. Evaluated on the SHWD benchmark, Rep-YOLO11s achieves 94.83% mAP_{50} with a pure GPU inference latency of 2.14 ms (467.3 FPS) on TensorRT FP16, while maintaining an end-to-end RTSP pipeline throughput of 65–95 FPS.

Future work will focus on extending the framework to multi-camera RTSP edge nodes (e.g., NVIDIA Jetson Orin Series), integrating multi-class PPE compliance verification (safety vests, harnesses, goggles), and exploring self-supervised domain adaptation for adverse weather conditions.

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